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Instructions to Convert a B5.9G to a B Gas Plus and a B5.9LPG to a B LPG Plus

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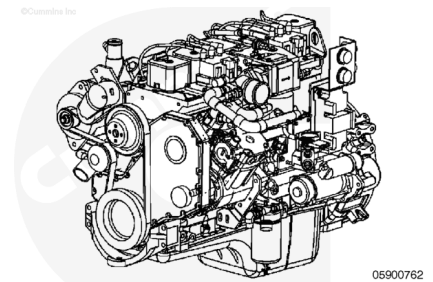
Preparatory Steps

The B Gas Plus and B LPG Plus engines require the use of an ISB (STORM) cylinder block. B5.9G and B5.9LPG engines built after March of 1997 were built with this cylinder block. The type of fitting in the oil rifle can identify the cylinder block. The correct fitting is a M14 straight thread o-ring. The older block has pipe threads.

After the completion of this upfit, the engine dataplate **must** be stamped with the new CPL. This engine modification **must** be reported to Cummins Warranty using Cummins Form 1877 (Notice of Engine Modification), which can be found in the Warranty Administration Manual.

Remove the following components:

- Turbocharger and exhaust adaptor
- Intake manifold cover and throttle assembly
- Fuel control housing
- Electronic control module (ECM)
- Ignition control module (ICM)
- Engine wiring harness.

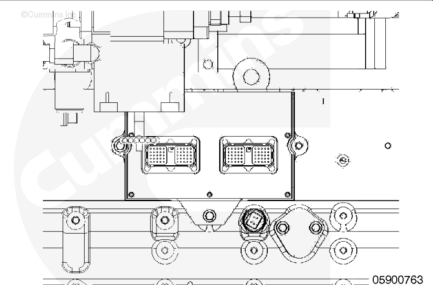


ECM

Attach the ECM, Part Number 3937299, to the cylinder block using spacers, Part Number 3936236, under the left and right ears. The center tab and left ear are held down with capscrew, Part Number 3900631. The right ear is held with capscrew, Part Number 3909025.

Note : The right ear capscrew is common with the ignition control module (ICM) bracket.

Replace the stud in the lift pump block off plate with an M8x20 capscrew, Part Number 3900630 (natural gas **only**).



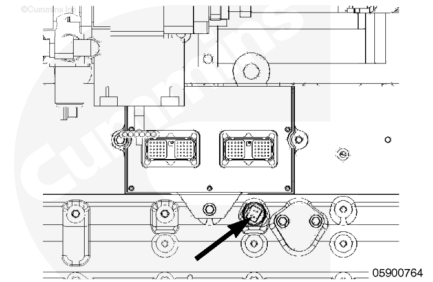
For B LPG Plus, install lift pump block-off plate, Part Number 3948083, and gasket, Part Number 3939258. Mount with two M8x20 capscrews, Part Number 3900630.

Refer to Procedure 019-031 in Troubleshooting and Repair Manual - Electronics, Gas Plus Engines, Bulletin 4021317.

Oil Pressure/Temperature Sensor

Insert the oil pressure/temperature sensor, Part Number 3417195, in the threaded hole below the ECM and to the left of the lift pump block off plate.

Refer to Procedure 019-155 in Troubleshooting and Repair Manual - Electronics, Gas Plus Engines, Bulletin 4021317.



Fuel Control Housing

Remove the gas mass flow sensor and the fuel shutoff valve (natural gas **only**) from the original housing.

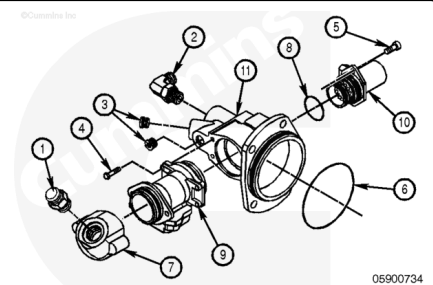
Install a new screen pack, Part Number 3929159, and the shutoff valve (natural gas **only**) or cover plate, Part Number 3964363, (LPG **only**) in the base of the new fuel control housing, Part Number 3938872.

Install a 45-degree connector, Part Number 200293, in the fuel inlet port (2).

The gas mass flow sensor housing is held in place with four M5 capscrews, Part Number 3253888, from the top rather than through-bolted.

Replace the 90-degree fuel connection fitting on the top of the gas mass flow sensor adaptor with a straight fitting, Part Number 129866.

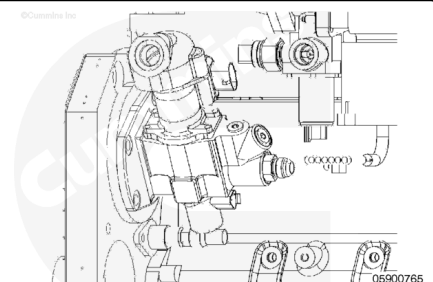
Refer to Procedure 005-009 in Troubleshooting and Repair Manual, B Series - Natural Gas, Bulletin 3666164.



Install a new o-ring on the fuel control housing.

Attach the fuel control housing with the same nuts.

Refer to Procedure 005-009 in Troubleshooting and Repair Manual, B Series - Natural Gas, Bulletin 3666164.



Ignition Module Bracket

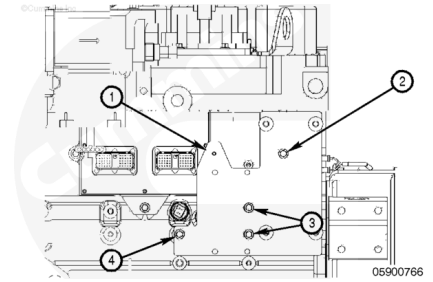
Attach the ignition module bracket, Part Number 3938875.

The bracket is held with one M8x55 capscrew (1), Part Number 3909025, one M8x80 capscrew (2), Part Number 3902112, two M8x35 capscrews (3), Part Number 3900632, and one M8xM6 stud (4), Part Number 3923022.

Note : Stud, Part Number 3909025, is common with the right ECM mounting tab.

Note : Capscrew, Part Number 3923022, is mounted in the lower left corner of the ICM bracket. The stud is used to mount the engine wiring harness ground terminal. Use M6 nut, Part Number 3906216.

Note : If a left side starter option is used with the starter solenoid on the top, there can be an interference problem with the low pressure regulator or evaporator. The starter option will need to be changed.



Fuel Pressure Regulator Housing (Natural Gas Only)

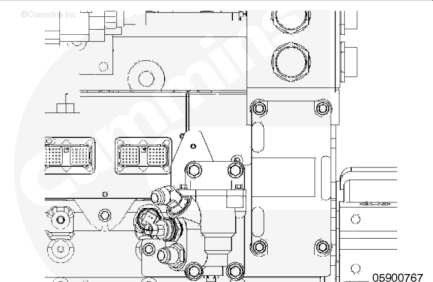
The pressure regulator housing, Part Number 3938679, attaches to the bracket with 1-M10x95 capscrew, Part Number 3908226, and 3-M10x25 capscrews, Part Number 3902460.

Orient the pressure regulator housing so the fittings are facing to the left (forward).

Note : The fuel outlet fitting (lower) may need to be replaced with a 90-degree fitting, Part Number 129859, if not already installed.

Note : Fuel pressure sensor primary, Part Number 3330527, is included in the pressure regulator assembly.

Refer to Procedure 005-047 in Troubleshooting and Repair Manual, B Series - Natural Gas, Bulletin 3666164.

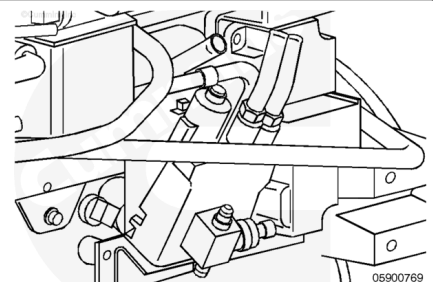


Fuel Evaporator Assembly (LPG Only)

The evaporator assembly, Part Number 3964356, attaches to the ignition module bracket with 4-M10x25 capscrews, Part Number 3902460.

Note : Fuel pressure sensor primary, Part Number 3330527, is included in the evaporator assembly.

Replace the two coolant tubes with Part Numbers 3967820 and 3967822.



Refer to Procedure 005-094 in Troubleshooting and Repair Manual, B Series - Natural Gas, Bulletin 3666164.

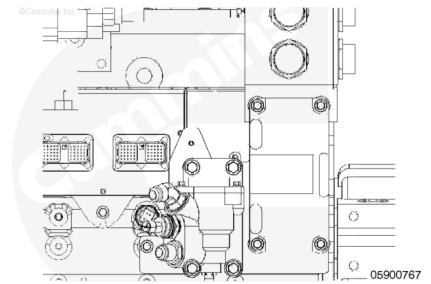
Note : The coolant line that is connected at the thermostat housing connects to the inboard fitting on the evaporator.

Ignition Module/OEM Connector Bracket

Attach the ignition module to the ignition module bracket. The module should be oriented with the two connectors for the ignition coils on the left side (forward). The OEM connector bracket, Part Number 3938921, is mounted behind the top two ICM mounting bolts.

Use 4-M8x40, Part Number 3900678, capscrews.

Refer to Procedure 019-105 in Troubleshooting and Repair Manual - Electronics, Gas Plus Engines, Bulletin 4021317.



Engine Wiring Harness

Install the engine wiring harness, Part Number 3938919. The ECM connectors are keyed to allow for correct orientation.

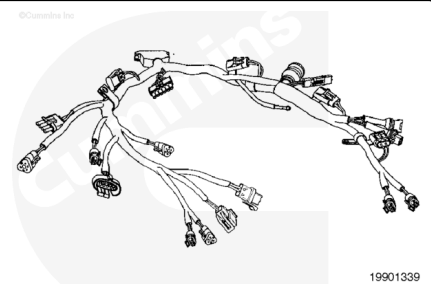
Mount the OEM connectors in the OEM connector bracket. Use washers, part Number 3617959, and nuts, Part Number 3617958, to secure the connectors.

Connect the ground terminal to the lower left ICM bracket stud with a M6 nut, Part Number 3906216.

Connect the two p-clips. The front p-clip is mounted to the upper outside fuel housing bolt. The second is mounted to the block behind the air compressor with capscrew, Part Number 3900629. It is mounted in the upper mounting hole.

Connect the engine wiring harness to the sensors and actuators as they are installed on the engine.

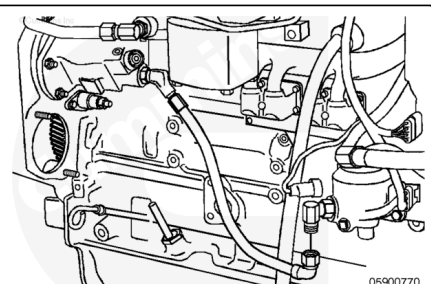
Refer to Procedure 019-043 in Troubleshooting and Repair Manual - Electronics, Gas Plus Engines, Bulletin 4021317.



Fuel Supply Hose

Connect the fuel supply hose to the low pressure regulator housing or evaporator and the fuel control housing, Part Number 3930037.

Note : The 90-degree end of the hose connects to the low pressure regulator housing.



Intake Manifold Cover

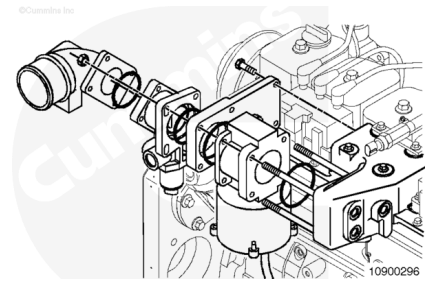
The throttle assembly can be attached to the intake manifold cover on or off the engine.

Insert the threaded rods, Part Number 3992127, into the intake manifold cover, Part Number 3938881.

Note : Install the four o-rings, Part Number 145582, between each mating surface.

Attach the throttle actuator, Part Number 3992126, to the intake manifold.

Attach the throttle support bracket, Part Number 3992124, to the throttle actuator and the intake manifold cover; use three tapered capscrews, Part Number 3284216.



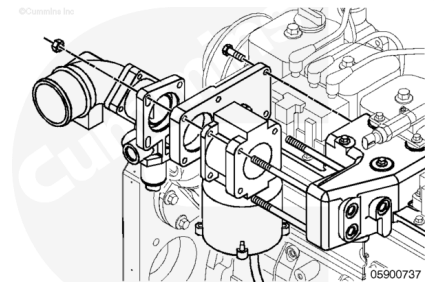
Slide the mixer housing, Part Number 3938871, onto the threaded rods.

Install the four nuts, Part Number 3900589.

Install intake elbow, Part Number 3938918, using the original intake elbow hardware.

The new intake elbow has the provision for a humidity sensor. The elbow can be oriented straight up or straight out.

Refer to Procedure 005-052 in Troubleshooting and Repair Manual, B Series - Natural Gas, Bulletin 3666164.



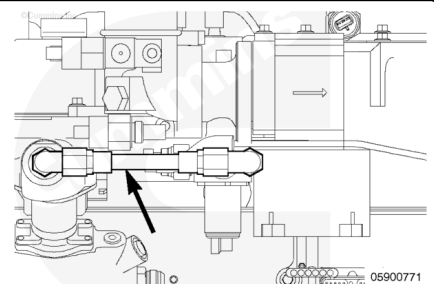
Fuel Jumper Hose

Install a 90-degree fitting, Part Number 129859, in the fuel mixer housing.

Install fuel control valve, Part Number 3938841-natural gas or Part Number 3938870-LPG, into the fuel mixer housing.

Connect the fuel jumper house, Part Number 3938932, to the 90-degree fitting on the mixer housing. If the intake/throttle assembly is being assembled on the engine, connect the hose to the gas mass flow sensor adaptor. Otherwise, make the connection after the intake manifold assembly is installed on the engine.

Install the ignition coils after the intake manifold is installed on the engine. Refer to Procedure 013-012 in Troubleshooting and Repair Manual, B Series - Natural Gas, Bulletin 3666164.

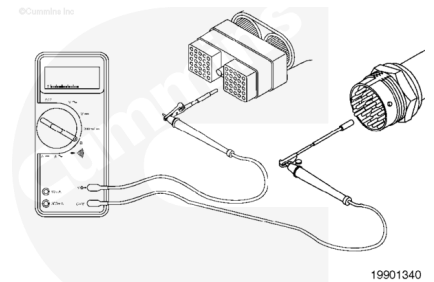


Refer to Procedure 005-052 in Troubleshooting and Repair Manual, B Series - Natural Gas, Bulletin 3666164.

Fuel Pressure/Temperature Sensor, Secondary

Install the fuel pressure/temperature sensor, Part Number 3417195, into the mixer housing.

Refer to Procedure 019-053 in Troubleshooting and Repair Manual - Electronics, Gas Plus Engines, Bulletin 4021317.

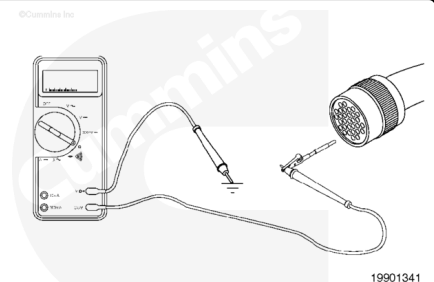


Intake Manifold Pressure/Temperature Sensor

Install the intake manifold pressure/temperature sensor, Part Number 4009913, into the intake manifold.

Install a plug in the remaining two holes in the intake manifold.

Refer to Procedure 019-099 in Troubleshooting and Repair Manual - Electronics, Gas Plus Engines, Bulletin 4021317.



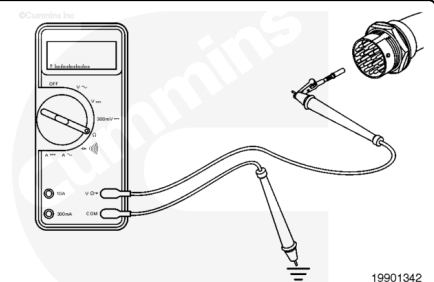
Knock Sensors

The engine is equipped with two knock sensors, Part Number 3607945.

The front sensor is located on the top of the intake manifold cover and shares a mounting bolt with the front inner bolt. A longer capscrew is required to mount the sensor, M8x45, part Number 3901445.

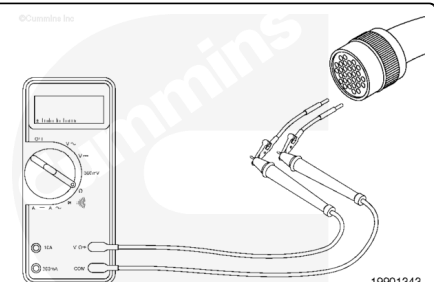
The rear sensor is located below the ignition module bracket. There is a boss on the engine with two threaded holes. The sensor is located in the lower hole and is mounted using knock sensor stud, Part Number 3992125, and M8 nut, Part Number 3900589.

Refer to Procedure 019-346 in Troubleshooting and Repair Manual - Electronics, Gas Plus Engines, Bulletin 4021317.



Exhaust Back Pressure Sensor

The exhaust back pressure sensor, Part Number 3348579, is remote mounted on the top of the intake manifold in sensor adaptor, Part Number 3938873. The sensor bracket is mounted with capscrews, Part Number 3937181.



The sensor connects to the exhaust adaptor through a stainless steel tube. The tube part number varies by turbocharger configuration.

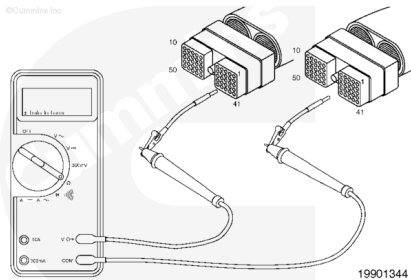
Refer to the turbocharger section for tube selection details.

Refer to Procedure 019-347 in Troubleshooting and Repair Manual - Electronics, Gas Plus Engines, Bulletin 4021317.

Ignition Harness

Connect the ignition harness, Part Number 3938920, to the ignition module and to both ignition coils.

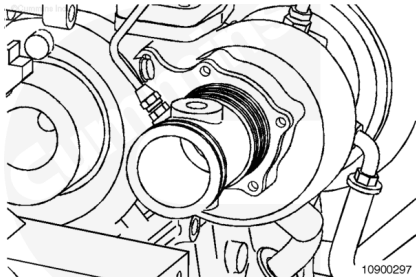
Refer to Procedure 013-037 in Troubleshooting and Repair Manual, B Series - Natural Gas, Bulletin 3666164.



Turbocharger

The B Gas Plus has several turbocharger configuration options. The turbocharger and exhaust adaptor will need to be replaced along with the installation of the exhaust back pressure sensor tube.

Refer to the following chart for option details.



Exhaust Back Pressure Tube Compatibility Chart

		Exhaust Adaptor		
		90-Degree Elbow	90-Degree Elbow (Rotatable)	Straight Out
Exhaust Manifold		XS 9260	XS 9266	XS 9259
LMRO	TB 9795	N/A	PH 9051	PH 9843
LMFO	TB 97003	PH 9844	N/A	N/A

Option Part Number List			
Option Number	Option Part Number	Required Fitting	Turbocharger
XS 9260	3992179	68138	N/A
XS 9266	3992179	3992178	N/A
XS 9259	3938917	68138	N/A
PH 9844	3938929	N/A	N/A
PH 9051	3992224	N/A	N/A
PH 9843	3938926	N/A	N/A
TB 9795	N/A	N/A	3599492
TB 97003	N/A	N/A	3599493

Conversion Parts List

Part Number	Part Description	Quantity	NG	LPG
68183	3/8 inch to 1/4 inch NPT	1	Yes	Yes
129859	90-degree fitting	1	Yes	No
129866	Straight fitting	1	Yes	Yes
145582	Throttle/mixer o-rings (4 total)	4	Yes	Yes
200293	45-degree fitting	1	Yes	Yes
3253888	GMFS mounting bolts M5	4	Yes	Yes
3284216	Throttle actuator/mixer bracket screws	3	yes	Yes
3348579	Back pressure sensor	1	Yes	Yes
3417195	Fuel pressure oil pressure combo sensor	2	Yes	Yes
3599492	Turbocharger LMRO	1 of 2	Yes	Yes
3599493	Turbocharger LMFO	1 of 2	Yes	Yes
3607945	Knock sensor	2	Yes	Yes

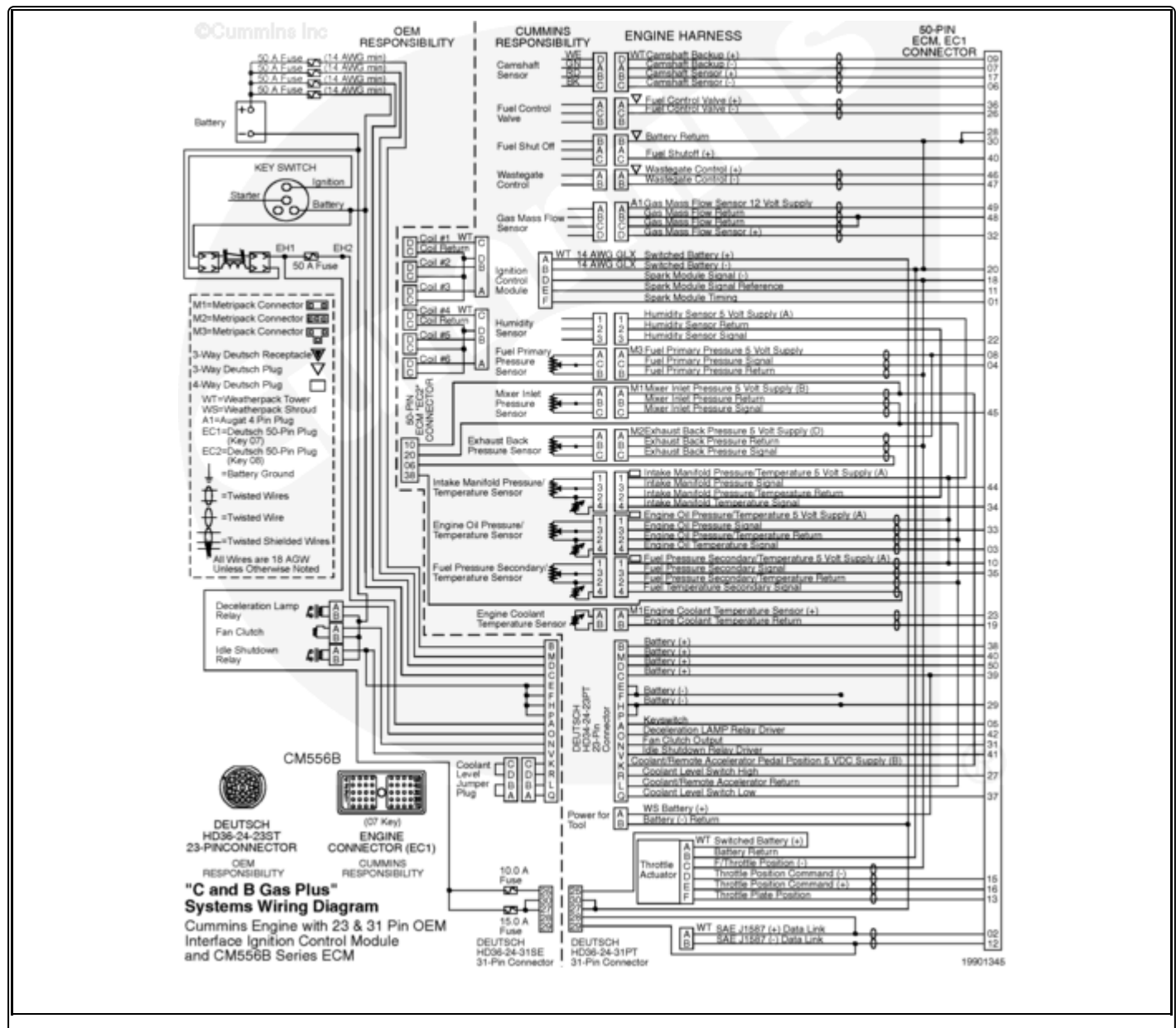
Part Number	Part Description	Quantity	NG	LPG
3617959	OEM connector washer	2	Yes	Yes
3900589	Nuts - fuel mixer M8	5	Yes	Yes
3925883	Engine harness p-clip	1	Yes	Yes
3900631	Holds-holds ECM - M8x25	4	Yes	Yes
3900632	Holds-holds IM bracket - M8x35	2	Yes	Yes
3900678	Holds-holds IM - M8x40	4	Yes	Yes
3901445	Capscrew M8x45	1	Yes	Yes
3902112	Holds-holds IM bracket - M8x80	1	Yes	Yes
3902460	Capscrew-holds pressure regulator M10x25	3	Yes	No
3902460	Capscrew M10x25 - evaporator mount	4	No	Yes
3906216	Nut - engine harness ground M6	1	Yes	Yes
3908226	Holds-holds pressure regulator M10x95	1	Yes	No
3909025	Holds-holds ECM/IM bracket M8x55	1	Yes	Yes
3923022	Stud - ICM mounting/engine harness ground	1	Yes	Yes
3929159	Screen pack	1	Yes	Yes

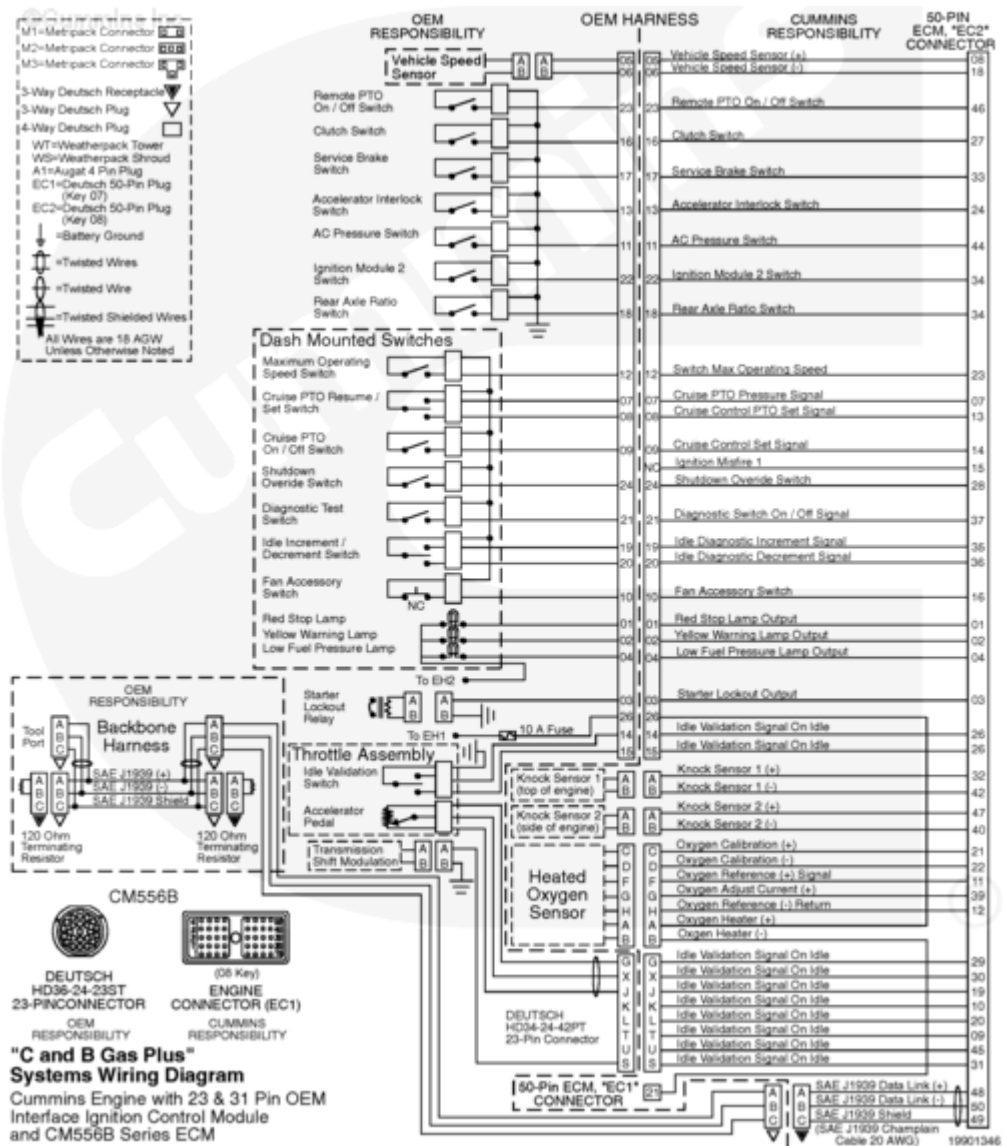
Part Number	Part Description	Quantity	NG	LPG
3930037	Fuel hose (pressure regulator to fuel control housing)	1	Yes	No
3936236	ECM holders	2	Yes	Yes
3937181	Capscrew - exhaust back pressure mounting	2	Yes	Yes
3937299	ECM	1	Yes	Yes
3938679	Pressure regulator assembly	1	Yes	No
3938836	Fuel control valve	1	No	Yes
3938841	Fuel control valve	1	Yes	No
3938871	Air fuel mixer	1	Yes	Yes
3938872	Fuel control housing	1	Yes	Yes
3938873	Back pressure sensor holder	1	Yes	Yes
3938875	Ignition module bracket	1	Yes	Yes
3938881	Intake manifold cover	1	Yes	Yes
3938917	Exhaust adaptor, straight	1 of 2	Yes	Yes
3938918	Air intake elbow	1	Yes	Yes
3938919	Engine wiring harness	1	Yes	Yes
3938920	Ignition system wiring harness	1	Yes	Yes
3938921	OEM connector bracket	1	Yes	Yes
3938926	Exhaust pressure sensing tube- LMRO straight	1 of 3	Yes	Yes

Part Number	Part Description	Quantity	NG	LPG
3938929	Exhaust pressure sensing tube-BBUK	1 of 3	Yes	Yes
3938932	Fuel supply hose (fuel housing to mixer)	1	Yes	Yes
3964356	Evaporator assembly	1	No	Yes
3964363	Fuel shutoff valve cover plate	1	No	Yes
3992124	Throttle support bracket	1	Yes	Yes
3992125	Knock sensor stud	1	Yes	Yes
3992126	Woodward throttle actuator	1	Yes	Yes
3992127	Long fuel system studs	4	Yes	Yes
3992178	Rear out adapter for elbow	1	Yes	Yes
3992179	Exhaust adaptor, 90-degree elbow	1 of 2	Yes	Yes
3992224	Pressure sensing tube-LMRO 90 rotatable	1 of 3	Yes	Yes
4001675	NTK heated oxygen sensor	1	Yes	Yes
4009913	Intake manifold combo sensor	1	Yes	Yes
3617958	OEM connector nut	2	Yes	Yes
4062315	Humidity sensor	1	Yes	Yes
3967820	LPG Evaporator coolant tube	1	No	Yes
3967822	LPG Evaporator coolant tube	1	No	Yes

OEM Installation Wiring Diagram

Note : You can also reference service wiring diagram, Bulletin 4021276.





Reference Documentation

Bulletin Number	Description
4021390	B5.9G, B5.9LPG, B Gas Plus, and B LPG Plus Owner's Manual
3666164	Troubleshooting and Repair Manual, B5.9G (Natural Gas) and B5.9LPG (Liquefied Petroleum Gas) Engines
4056515	B Gas Plus Parts Catalog
4021317	Troubleshooting and Repair Manual - Electronics, Gas Plus Engines
4021276	Gas Plus Wiring Diagram
3666119	B5.9G/B5.9LPG Wiring Diagram

For additional installation information, contact your local Cummins Authorized Repair Facility.

Document History

Date	Details
2017-03-27	Module Converted
2017-4-4	Update to Format

Last Modified: 04-Apr-2017
